

GATE SOLUTION IN MINING ENGINEERING

(Containing Solved Questions 2015 - 2007)

By
RUPESH KUMAR SAHU

LOVELY PRAKASHAN, DHANBAD

Gate Solution – 2007

Q . 1 If the slope of the diagonal of rectangular is m , the slope of the other diagonal is

- (A) $\frac{1}{2m}$ (B) $-\frac{1}{2m}$ (C) $\frac{1}{m}$ (D) $-\frac{1}{m}$

Answer (D)

Solution . We know,

Product of slope of diagonals of rectangular is equal to (-1) .

So,

$$m_1 \times m_2 = -1$$

Given that, $m_1 = m$

$$m \times m_2 = -1$$

$$\therefore m_2 = \frac{-1}{m}$$

Q .2 if the rank of the matrix A is r , the rank of the matrix A^T is

- (A) r , if and only if $A^T = A$ (B) r , for all A
(C) p , where $p \neq r$ (D) $r - 1$, where $r \geq 1$

Answer (D)

Solution .

Suppose, a matrix $(A) = \begin{bmatrix} 1 & 2 & 1 \\ 3 & 2 & 1 \\ 1 & 1 & 2 \end{bmatrix}$

Thus, rank of $(A) = 3$, and

Transpose matrix of A matrix is given by-

$$A^T = \begin{bmatrix} 1 & 3 & 1 \\ 2 & 2 & 1 \\ 1 & 1 & 2 \end{bmatrix}$$

Thus, rank of $(A^T) = 2$

$\therefore r-1$, where $r \geq 1$

Q.3 Bulk modulus of rock defined as

(A) $\frac{\text{shear stress}}{\text{volumetric strain}}$

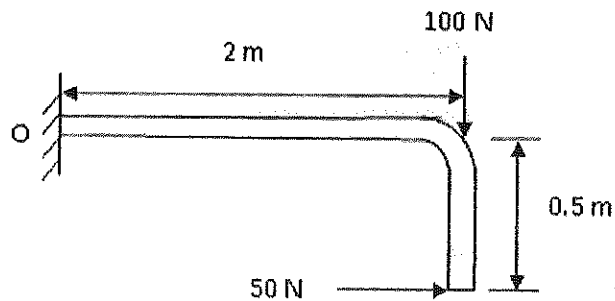
(B) $\frac{\text{hydrostatic pressure}}{\text{shear strain}}$

(C) $\frac{\text{hydrostatic pressure}}{\text{volumetric strain}}$

(D) $\frac{\text{shear stress}}{\text{shear strain}}$

Answer (C)

Q.4 The magnitude of the resultant moment about point O in Nm of two forces acting on the rod shown below is



(A) 25

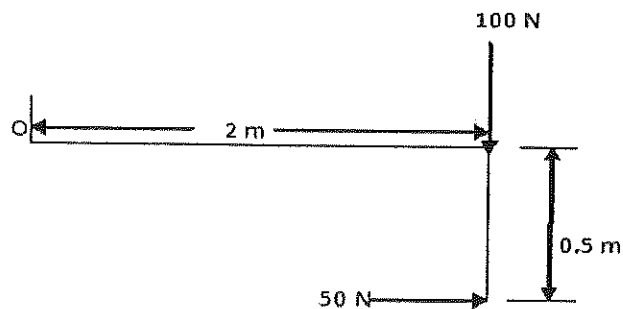
(B) 125

(C) 175

(D) 225

Answer (C)

Solution .



Magnitude of resultant moment about O :-

$$M_O = 100 \times 2 - 50 \times 0.5$$

$$M_0 = 175 \text{ kN.m}$$

Note - clock wise moment is taken (+) and anti clock wise (-)

Q 5. Radial stress on the excavation boundary of circular tunnel is

- (A) always zero
- (B) always positive
- (C) always negative
- (D) positive in some area and negative in some area

Answer (A)

Q .6 The critical diameter of an explosive is defined as the diameter below which it

- (A) develops the optimum velocity of detonation
- (B) does not involve in chemical reaction
- (C) develops the maximum velocity of detonation
- (D) deflagrates

Answer (D)

Q.7 Which one of the following supports does NOT require a power pack for its operation

- (A) chock shield support
- (B) open circuit hydraulic prop
- (C) close circuit hydraulic prop
- (D) Alpine breaker line support

Answer (C)

Q 8. In a centrifugal flow fan the conversion of velocity pressure to static pressure accomplished

with the help of

(A) impeller

(B) curved blades

(C) hub

(D) casing

Answer (D)

Q 9. A 3.3 kV, 3-phase AC motor having a PF of 0.85 draws current at 95 A . The motor input power in kW is

(A) 266.5

(B) 461.5

(C) 543

(D) 799.5

Answer (B)

Solution :-

For three phase supply , motor input power is given by –

$$P = \sqrt{3} \cdot V_L \cdot I_L \cdot \cos\phi$$

$$P = \sqrt{3} \times 3.3 \times 10^3 \times 95 \times 0.85$$

$$P = 461548 \text{ W}$$

$$P = 461.5 \text{ KW}$$

Q 10. The amount of total stone dust required in kg for a secondary heavy type stone dust barrier in a roadway of size $4.0 \text{ m} \times 3.0 \text{ m}$ is

(A) 1320

(B) 4680

(C) 5200

(D) 6600

Answer (B)

Solution .

For heavy type or secondary barrier, stone dust required per m^2 is 390 kg.

So, Total stone dust required in roadway of size $(4.0 \times 3.0) = 12 \times 390 \text{ kg} = 4680 \text{ kg}$.

Q 11. In the Gaussian plume model, the dispersion coefficient are function of

- (A) distance from the source and stability class
- (B) stack height and distance from the source
- (C) stability class and source coordinates
- (D) source coordinates and distance from the source

Answer (A)

Q 12. The ratchet-and- pawl arrangement in percussive drill machine helps in

- (A) providing required rotational speed
- (B) indexing at the bit rock interface
- (C) regulating airflow in forward and return strokes of the piston
- (D) engaging the bit with the rock between the blows

Answer (B)

Q 13. The measurement of distances from a position on the earth to artificial satellites is known as

- (A) astronomical ranging (B) pseudo ranging
- (C) satellite ranging (D) celestial ranging

Answer (B)

Q 14. In opencast mining, the width which is extracted from the working bench is termed as

- (A) cut (B) bench width (C) bank width (D) bench face

Answer (A)

Q 15. Zener barriers are associated with

- (A) increased safety apparatus
- (B) statistically safe apparatus

(C) flame proof apparatus

(D) intrinsic safety apparatus

Answer (D)

Q 16. The most recent model of self- contained compressed oxygen breathing apparatus is

(A) Proto-IV

(B) BG-174

(C) BG-4

(D) BG-174 A

Answer (C)

Q 17. The measure of dispersion are

(A) range, variance and standard deviation

(B) mean, median and variance

(C) mean, mode and skewness

(D) mean, range and variance

Answer (A)

Q 18. In a single server queueing model with constant arrival rate, which one of the following probability distribution is followed by the inter-arrival times of the costumers at the service facility?

(A) binomial

(B) Poisson

(C) Weibull

(D) exponential

Answer (B)

Q 19. A company invested Rs. 4 lakh in a machine with an expected useful life of 12 years. The net income expected from the operation of the machine is Rs. 80,000 per annum. The payback period of the machine in years is

(A) 4

(B) 5

(C) 6

(D) 7

Answer (B)

Solution .

Machine earned per year = 80,000 rupees.

So, time (T) required to pay 4,00,000 rupees = $\frac{400000}{80000}$, year

T = 5 years.

Q 20. The angular (horizontal/vertical) observation made by a transit theodolite with the face of the vertical circle on the right side of the observer is called

- (A) face right observation (B) face left observation
(C) normal observation (D) reciprocal observation

Answer (A)

Q 21. Two sides of a triangle are represented by vectors $\vec{a} = \hat{i} + \hat{j} + \hat{k}$ and $\vec{b} = -\hat{i} - \hat{j} + \hat{k}$. The area (magnitude) of the triangle is

- (A) $1/\sqrt{2}$ (B) 1 (C) $\sqrt{2}$ (D) $2\sqrt{2}$

Answer (C)

Solution .

$$\text{Let, } \vec{a} = \hat{i} + \hat{j} + \hat{k}$$

$$\vec{b} = -\hat{i} - \hat{j} + \hat{k}$$

$$\text{Now, } \vec{a} \times \vec{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 1 & 1 \\ -1 & -1 & 1 \end{vmatrix}$$

$$\vec{a} \times \vec{b} = \hat{i}(1 + 1) - \hat{j}(1 + 1) + \hat{k}(-1 + 1)$$

$$\vec{a} \times \vec{b} = 2\hat{i} - 2\hat{j}$$

$$\therefore \text{Area of triangle } (A_T) = \frac{1}{2} |\vec{a} \times \vec{b}|$$

$$A_T = \frac{1}{2} |2\hat{i} - 2\hat{j}|$$

$$\therefore A_T = \frac{1}{2} \sqrt{4 + 4}$$

$$\therefore A_T = \sqrt{2}$$

Q 22. The cost of diesel is Rs. $\left(25 + \frac{x}{90}\right)$ per km to drive a dump truck at a speed of x km/hour.

The maintenance cost of the truck is Rs. 10 per hour. To minimize the cost per km, the truck speed in km/hour is

- (A) 5 (B) 20 (C) 25 (D) 30

Answer (D)

Solution .

Cost of diesel = Rs. $\left(25 + \frac{x}{90}\right)$ per km.

Maintenance cost = Rs. 10 per hour

Or maintenance cost = Rs. $\frac{10}{x}$ per Km

So, total cost (c) = $\left[\left(25 + \frac{x}{90}\right) + \frac{10}{x} \right]$ per Km

Now,

$$\frac{dc}{dx} = \frac{1}{90} - \frac{10}{x^2}$$

For minimum or maximum, $\frac{dc}{dx} = 0$.

$$\frac{1}{90} - \frac{10}{x^2} = 0$$

$$\frac{10}{x^2} = \frac{1}{90}$$

$$x^2 = 900$$

$$x = \sqrt{900} = 30 \text{ Km per hour}$$

Q 23. The function $f(x)$ and $g(x)$ satisfy $f(x=0) = 3$, $f'(x=0) = -5$, $g(x=0) = 2$ and

$g'(x = 0) = -10$. The value of $\frac{d}{dx} \left(\frac{f(x)}{g(x)} \right)_{x=0}$ is

- (A) -35.0 (B) -5.0 (C) 0.5 (D) 5.0

Answer (D)

Solution .

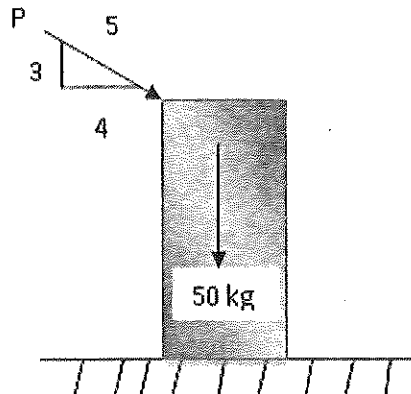
Given that, $f(x = 0) = 3$, $f'(x = 0) = -5$, $g(x = 0) = 2$ and $g'(x = 0) = -10$

The value of –

$$\begin{aligned} \frac{d}{dx} \left\{ \frac{f(x)}{g(x)} \right\}_{x=0} &= \frac{\frac{d}{dx} f(x) \cdot g(x) - \frac{d}{dx} g(x) \cdot f(x)}{\{g(x)\}^2} \\ &= \frac{f'(x) \cdot g(x) - g'(x) \cdot f(x)}{\{g(x)\}^2} \\ &= \frac{-5 \times 2 - (-10) \times 3}{2^2} \\ &= \frac{-10 + 30}{4} \end{aligned}$$

$$\frac{d}{dx} \left\{ \frac{f(x)}{g(x)} \right\}_{x=0} = 5$$

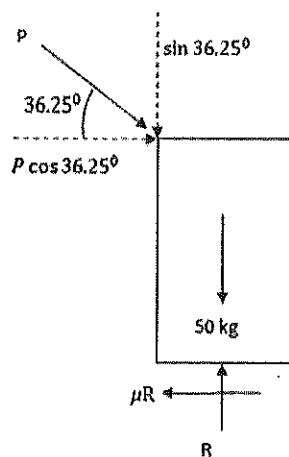
- Q.24 A wooden block of 50 kg rests on the floor (as shown in the figure below) for which the coefficient of friction is 0.5. The smallest magnitude of the force P in kg that will cause impending motion of the block is



- (A) 50 (B) 40 (C) 30 (D) 25

Answer (A)

Solution .



Here ,

$$\tan \theta = \frac{3}{4}$$

$$\theta = \tan^{-1} \frac{3}{4}$$

$$\therefore \theta = 36.25^\circ$$

$$\text{Now, } \sum H = 0$$

$$P \cos 36.25^\circ - \mu R = 0$$

$$\therefore R = \frac{P \cos 36.25^\circ}{\mu} = \frac{P \cos 36.25^\circ}{0.5} = 1.6 P$$

$$\text{And } \sum V = 0$$

$$-P \sin 36.25^\circ + R - 50 = 0$$

$$-0.6 P + 1.6 P - 50 = 0$$

$$P (1.6 - 0.6) = 50$$

$$\therefore P = 50 \text{ kg}$$

Q 25. The solution of $ye^x dx + (4y + e^x)dy = 0$ for $y(0) = -1$ is

(A) $ye^x + 2y^2 - 1 = 0$ (B) $e^x + xy^2 - 2 = 0$

(C) $ye^x - y^2 = 0$

(D) $xe^x + y^2 - 1 = 0$

Answer (A)

Solution .

Given that,

$$ye^x \cdot dx + (4y + e^x)dy = 0, y(0) = -1$$

$$\text{Here, } M = ye^x, N = (4y + e^x)$$

$$\therefore \frac{\partial M}{\partial y} = e^x, \quad \frac{\partial N}{\partial x} = e^x$$

It is seen that,

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x} \text{ and solution will be}$$

$$\int M(y - \text{constant}) dx + \int N(\text{term of } N \text{ not containing } x) dy = C$$

$$\int ye^x dx + \int 4y dy = C$$

$$ye^x + 4\frac{y^2}{2} = C \dots\dots\dots(1)$$

putting $y = -1, x = 0$ in equation (1)

$$\therefore C = 1$$

Therefore from equation (1), solution will be

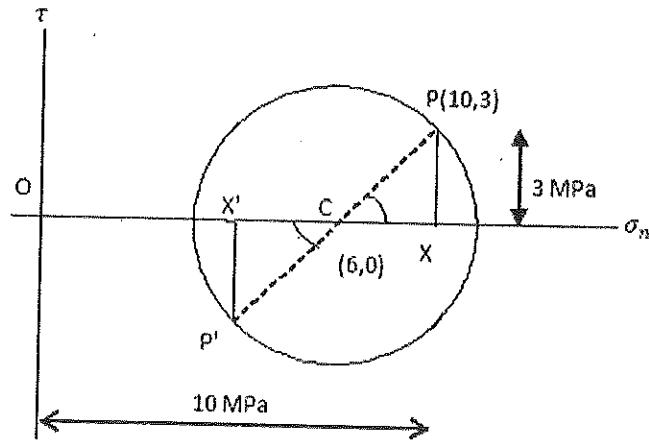
$$ye^x + 2y^2 - 1 = 0$$

Q 26. A point P (10,3) MPa on the Mohr's circle represents normal and shear stresses. If the centre of the Mohr's circle is C (6,0) MPa, the normal and shear stresses in MPa on the point diametrically opposite to P are

- (A) (2,-3) (B) (4,-3) (C) (2, 3) (D) (4, 3)

Answer (A)

Solution .



Here , $PX = 3 \text{ MPa}$, $OX = 10 \text{ MPa}$, $OC = 6 \text{ MPa}$

$\therefore P'X' = -3 \text{ MPa}$, (due to below of $OX'X$ line)

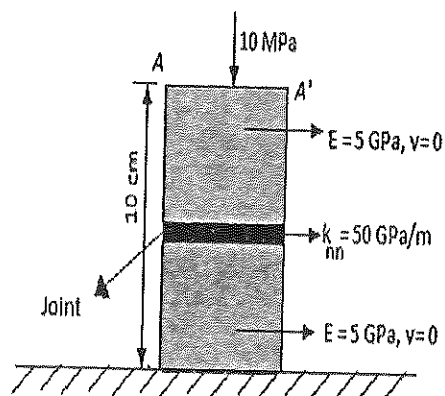
$$CX = 10 - 6 = 4 \text{ MPa}$$

$$\text{Also } CX' = 4 \text{ MPa}$$

$$\therefore OX' = 6 - 4 = 2 \text{ MPa}$$

So, normal and shear stresses on the point diametrically opposite to P is $(2, -3)$.

- Q 27.** A rock sample with a horizontal joint is subjected to 10 MPa of normal pressure as shown in the figure. The elastic modulus and Poisson ratio of the rock are 0.5 GPa and 0 respectively. If the normal stiffness (k_{nn}) of the joint is 50 GPa/m, normal displacement of the top of the sample (AA' line) in mm is



(A) 0.2

(B) 0.4

(C) 0.6

(D) 0.8

Answer (B)

solution .

For rock sample , applying Hook's law-

$$\sigma = E. \varepsilon$$

$$\text{Or } \sigma = E. \frac{\Delta L}{L}$$

$$10 \times 10^6 = \frac{5 \times 10^9 \times \Delta L}{10 \times 10^{-2}}$$

$$\Delta L = \frac{1}{5 \times 10^3}, m$$

$$\Delta L = 0.2 \text{ mm}$$

For joint,

$F = k.x$, where x = displacement and k = stiffness;

$$10 \times 10^6 = 50 \times 10^9 \times x$$

$$x = \frac{1}{5 \times 10^3}, m$$

$$x = 0.2 \text{ mm}$$

$$\therefore \text{Normal displacement (D)} = \Delta L + x$$

$$D = 0.2 + 0.2$$

$$D = 0.4 \text{ mm}$$

Q 28. The state of stress $(\sigma_{xx}, \sigma_{yy}, \tau_{xy})$ at a point below ground is found to be $(5, 15, -3)$ MPa.

The angle measured in the counter clockwise direction between the x -axis and major principal axis in degrees is

(A) 9.52

(B) 15.48

(C) 150.48

(D) 164.52

Answer (B)

Solution .

We have,

$$\tan 2\alpha = \frac{2 \tau_{xy}}{\sigma_{yy} - \sigma_{xx}}$$

$$\tan 2\alpha = \frac{-2 \times 3}{15 - 5}$$

$$\tan 2\alpha = -0.6$$

$$2\alpha = \tan^{-1}(-0.6)$$

$$\alpha = 15.48^\circ$$

Q 29. The unconfined compressive strength of a cylindrical rock sample is 90 MPa. The angle of internal friction of the rock is 30° . If a confining pressure of 5 MPa is applied radially to the rock sample, the confined compressive strength in MPa is

- (A) 92.88 (B) 95.00 (C) 105.00 (D) 110.0

Answer (C)

Solution .

Confined compressive strength (S) of rock sample is given by-

$$S = C_0 + \sigma_1 \tan \beta$$

Where, C_0 = uniaxial compressive strength = 90 MPa

σ_1 = confining pressure = 5 MPa, and

$$\tan \beta = \frac{1 + \sin \phi}{1 - \sin \phi}$$

[angle β constant, ϕ -angle of internal friction]

Therefore,

$$S = 90 + 3 \times 5$$

$$S = 105 \text{ MPa}$$

Q 30. A circular opening of radius a is made underground in hydrostatic condition. The radial

distance from the centre of the opening, where the tangential stress is twice the radial stress, is

- (A) a (B) $\sqrt{2}a$ (C) $\sqrt{3}a$ (D) $2\sqrt{3}a$

Answer (C)

Solution .

Radial stress (σ_r) is given by:-

$$\sigma_r = \frac{1}{2} P_z \left\{ (1+K) \left(1 - \frac{a^2}{r^2} \right) + (1-K) \left(1 - 4 \frac{a^2}{r^2} + 3 \frac{a^4}{r^4} \cos 2\theta \right) \right\}$$

Tangential stress is given by:

$$\sigma_\theta = \frac{1}{2} P_z \left\{ (1+K) \left(1 + \frac{a^2}{r^2} \right) + (1-K) \left(1 + 3 \frac{a^4}{r^4} \right) \cos 2\theta \right\}$$

In hydrostatic stress condition-

$$\sigma_x = \sigma_y; \quad K = \frac{\sigma_x}{\sigma_y} = 1;$$

But as per given condition:

$$\sigma_\theta = 2\sigma_r$$

$$\therefore \frac{1}{2} P_z \left\{ (1+K) \left(1 + \frac{a^2}{r^2} \right) + (1-K) \left(1 + 3 \frac{a^4}{r^4} \right) \cos 2\theta \right\} = 2 \cdot \frac{1}{2} P_z \left\{ (1+K) \left(1 - \frac{a^2}{r^2} \right) + (1-K) \left(1 - 4 \frac{a^2}{r^2} + 3 \frac{a^4}{r^4} \cos 2\theta \right) \right\};$$

Putting $K = 1$; we get

$$\frac{1}{2} P_z \left\{ (1+1) \left(1 + \frac{a^2}{r^2} \right) \right\} = 2 \cdot \frac{1}{2} P_z \left\{ (1+1) \left(1 - \frac{a^2}{r^2} \right) \right\}$$

$$\therefore \quad 1 + \frac{a^2}{r^2} = 2 - \frac{2a^2}{r^2}$$

$$\frac{3a^2}{r^2} = 1$$

$$3a^2 = r^2$$

$$\therefore \quad r = \sqrt{3} \cdot a$$

Q 31. Coal pillar strength is represented by $S = S_1 h^\alpha w^\beta$, where S_1 = insitu strength of the pillar, h = mining height, and w = pillar width. Two bord and pillar panels are developed in similar geological conditions at depth D_1 and D_2 with mining heights h_1 and h_2 respectively. If the gallery width and the pillar width in both the panels remain the same, the ratio of pillar safety factors, SF_1/SF_2 is

- (A) $\left(\frac{h_2}{h_1}\right)^\alpha \frac{D_1}{D_2}$ (B) $\left(\frac{h_2}{h_1}\right)^\alpha \frac{D_2}{D_1}$ (C) $\left(\frac{h_1}{h_2}\right)^\alpha \frac{D_1}{D_2}$ (D) $\left(\frac{h_1}{h_2}\right)^\alpha \frac{D_2}{D_1}$

Answer (D)

Solution.

For Panel – I

$$\text{Strength of pillar} = S_1 (h^\alpha)_1 W^\beta$$

$$\text{Load acting on pillar} = \gamma D_1 \frac{(a+b)^2}{a^2}$$

$$\therefore FS_1 = \frac{S_1 \cdot (h^\alpha)_1 \cdot W^\beta}{\gamma \cdot D_1 \cdot \frac{(a+b)^2}{a^2}} \dots\dots\dots(1)$$

For Panel – II

$$\text{Strength of pillar} = S_1 (h^\alpha)_2 W^\beta$$

$$\text{Load acting on pillar} = \gamma D_2 \frac{(a+b)^2}{a^2}$$

$$\therefore FS_2 = \frac{S_1 \cdot (h^\alpha)_2 \cdot W^\beta}{\gamma \cdot D_2 \cdot \frac{(a+b)^2}{a^2}} \dots\dots\dots(2)$$

From equation (1) & (2)

$$\frac{FS_1}{FS_2} = \frac{\frac{S_1 \cdot (h^\alpha)_1 \cdot W^\beta}{\gamma \cdot D_1 \cdot \frac{(a+b)^2}{a^2}}}{\frac{S_1 \cdot (h^\alpha)_2 \cdot W^\beta}{\gamma \cdot D_2 \cdot \frac{(a+b)^2}{a^2}}}$$

$$\frac{FS_1}{FS_2} = \left(\frac{h_1}{h_2}\right)^\alpha \cdot \frac{D_2}{D_1}$$

Q 32.

Match the following

Belt conveyor component

Function

P. Pull cord

1. Cleaning device

Q. Snub pulley

2. Discharging material on the side of the conveyor

R. Tripper

3. Safety stopping device

S. rotary brush

4. Increasing the angle of wrap

(A) P-1, Q-2, R-3, S-4

(B) P-3, Q-4, R-1, S-2

(C) P-4, Q-2, R-3, S-1

(D) P-3, Q-4, R-2, S-1

Answer (D)

Q 33.

Match the following

Equipment

Action /Purpose

P. Dragline

1.Reaming

Q. Bucket wheel excavator

2. Key cut

R. Tunnel Boring Machine

3. Pulsating impact

S. Hydraulic monitor

4. Terracing

(A) P-1, Q-2, R-3, S-4

(B) P-2, Q-4, R-1, S-3

(C)P-2, Q-4, R-3, S-1

(D) P-3, Q-4, R-2, S-1

Answer (B)

Q 34. Match the following

Mining method

Face supporting system

P. Mechanizwd longwall

1. Cable bolting

Q. Blasting gallery

2. Shield type powered support

R. Steep seam mechanized longwall

3. Alpine breaker line supports

S. Wangawilli

4. Troika shield supports

(A) P-1, Q-2, R-3, S-4

(B) P-2, Q-1, R-4, S-3

(C) P-3, Q-4, R-2, S-1

(D) P-2, Q-4, R-1, S-3

Answer (B)

Q 35. A 15 yd^3 dragline is deployed in an overburden bench of an opencast mine. It works for 40 days at the rate of 6 hours per shift and 3 shifts a day. The cycle time, bucket fill factor, and operating efficiency of the dragline are respectively 50 s, 0.8, and 75%. The total volume of overburden in m^3 handled by the dragline is

($1 \text{ yd}^3 = 0.765 \text{ m}^3$)

(A) 356918

(B) 634521

(C) 557685

(D) 991440

Answer (A)

Solution .

Total volume (A_m) handled by dragline is given by:-

$$A_m = \frac{3600 \times C_b \times F \times \alpha \times \eta}{t \times S}, \text{ m}^3/\text{h}$$

Where, C_b = size of bucket in $\text{m}^3 = 15 \times 0.765 = 11.475 \text{ m}^3$

F = fill factor = 0.8

t = cycle time in seconds = 50 seconds

η = operating efficiency = 0.75

$$A_m = \frac{3600 \times 11.475 \times 0.8 \times 0.75}{50}$$

$$A_m = 495.72 \text{ m}^3/\text{h}$$

$$A_m = 495.72 \times 40 \times 3 \times 6 = 356918 \text{ m}^3$$

Q 36. The phenomenon of fretting (necking) of pillar in room and pillar stoping is common in the pillars formed in

- (A) massive rock with very high pillar height to width ratio
- (B) regularly jointed rock with high pillar height to width ratio
- (C) massive rock with low pillar height to width ratio
- (D) transversely jointed rock with low pillar height to width ratio

Answer (B)

Q 37. In an underground opening , the immediate roof strata consists of two rock layers with the following properties:

Property	Layer-I	Layer-II
Modulus of elasticity (GPa)	60	40
Modulus of rupture (MPa)	20	10
Unit weight (kN/m ³)	25	20
Thickness (m)	5	2.5

Considering a factor of safety of 4.0, the length of safe span in m is

- (A) 27.82
- (B) 34.06
- (C) 36.54
- (D) 39.34

Answer (B)

Solution .

We have,

$$L = \sqrt{\frac{2 \cdot R_0 \cdot t}{\gamma \cdot F_s}}, \text{ m}$$

Where, L= length of the safe span in m

t = thickness in m

F_s = factor of safety = 4

γ = unit weight in kN/m^3

R_0 = modulus of rupture in MPa

$$\text{Here, } t = \frac{t_1 + t_2}{2} = \frac{5 + 2.5}{2} = 3.75 \text{ m}$$

$$\gamma = \frac{E_1 (t^2)_1 (\gamma_1 t_1 + \gamma_2 t_2)}{E_1 (t^3)_1 + E_2 (t^3)_2}$$

$$\gamma = \frac{60 \times 10^9 \times 5^2 [25 \times 10^3 \times 5 + 20 \times 10^3 \times 2.5]}{60 \times 10^9 \times 5^3 + 40 \times 10^9 \times 2.5^3}$$

$$\therefore \gamma = 32.30 \text{ kN/m}^3$$

And $R_0 = 20 \text{ MPa} = 20 \times 10^3 \text{ kPa}$

$$\therefore L = \sqrt{\frac{2 \times 20 \times 10^3 \times 3.75}{32.30 \times 4}}$$

$$\therefore L = 34.07 \text{ m}$$

Q 38. In an opencast mine, a centrifugal pump is required to lift water at a rate of 60 l/s to a height of 80 m above the pump level. The vertical suction head is 4 m. The total friction head including shock and energy loss is 10 m. If the pump runs at an efficiency of 80%, the brake power of motor in kW is

(A) 70.50

(B) 67.50

(C) 63.00

(D) 57.55

Answer (A)

Solution :-

$$\text{Input power to motor} = \frac{P \times Q \times 10^{-3}}{\text{pump efficiency}}, \text{ kW}$$

$$P = \text{pump pressure} = \rho \cdot g \cdot h = 1000 \times 9.81 \times (10 + 4 + 80) = 922140 \text{ Pa}$$

$$Q = 60 \text{ l/s} = 60 \times 10^{-3} \text{ m}^3/\text{s}$$

$$\therefore \text{Input power (P) to motor} = \frac{922140 \times 60 \times 10^{-3} \times 10^{-3}}{0.80} \text{ kW}$$

$$P = 69.2 \text{ kW}$$

Q 39. Match the following

Support system	support principle
P. shotcrete	1. Reinforces rock mass by binding them together
Q. Backfill	2. Acts as link between two layers of rock to transfer load between them
R. Bolt	3. Imposes kinematics constraints on key pieces in a stope boundary
S. Prop	4. Prevents spatially progressive disintegration of near field rock mass

(A) P-3, Q-4, R-2, S-1

(B) P-2, Q-1, R-4, S-2

(C) P-4, Q-3, R-1, S-2

(D) P-3, Q-4, R-1, S-2

Answer (C)

Q 40. Match the following

Stope	Drill machine	Method of drilling
P. Shrinkage	I. Drill jumbo	1. Fan drilling
Q. Room-and-pillar	J. Down-the-hole-hammer	2. Overhand drilling
R. Sublevel	K. Hand held stopper	3. Parallel drilling
S. Sublevel caving	L. Mechanized fan drill	4. Frontal vertical/downward benching

(A) P-I-2, Q-K-4, R-L-3, S-J-1

(B) P-K-4, Q-I-3, R-J-2, S-L-1

(C) P-K-2, Q-I-4, R-J-3, S-L-1

(D) P-I-3, Q-K-4, R-J-1, S-L-2

Answer (C)

Q41. A coal seam of 12 m thickness is worked out by mechanized top coal caving system. The thickness of the bottom slice is 3 m, length of solid coal face is 120 m and the average depth of cut by the shearer (web) is 70 cm. The density of coal is 1300 kg/m^3 with the percentage of extraction in the slice at 95 and in the top coal at 70. The production of coal per cycle in tonne is

- (A) 1008 (B) 999 (C) 688 (D) 311

Answer (B)

Solution .

$$\begin{aligned} \text{Production of coal in the slice} &= \frac{3 \times 120 \times 0.7 \times 0.95 \times 1300}{1000} \\ &= 311.22 \text{ tonnes} \end{aligned}$$

$$\begin{aligned} \text{And production of coal in the top} &= \frac{(12-3) \times 120 \times 0.7 \times 0.7 \times 1300}{1000} \\ &= 687.96 \text{ tonne} \end{aligned}$$

$$\begin{aligned} \therefore \text{Production of coal per cycle} &= 311.22 + 687.96 \\ &= 999.18 \text{ tonne} \end{aligned}$$

Q 42. Two reservoirs are connected by two equal length parallel pipe lines with diameter d and $2d$. assuming similar resistance coefficient, if the discharge through the smaller diameter pipeline is $0.04 \text{ m}^3/\text{s}$, the discharge through the other pipelines in m^3/s is

- (A) 0.226 (B) 0.426 (C) 1.130 (D) 1.280

Answer (A)

Solution.

We know,

$$\text{Pressure drop (P)} = \frac{K S Q^2}{A^3}$$

$$\text{Or } P = \frac{K (\pi D L) Q^2}{\left(\frac{\pi D^2}{4}\right)^3}$$

$$\therefore P = \frac{64 K L Q^2}{\pi^2 D^5}$$

If P, K & L constant, then

$$Q^2 \propto D^5$$

$$\text{Or } \frac{(Q^2)_1}{(Q^2)_2} = \frac{(D^5)_1}{(D^5)_2}$$

$$\text{Or } Q_2 = Q_1 \times \sqrt{\left(\frac{D_2}{D_1}\right)^5}$$

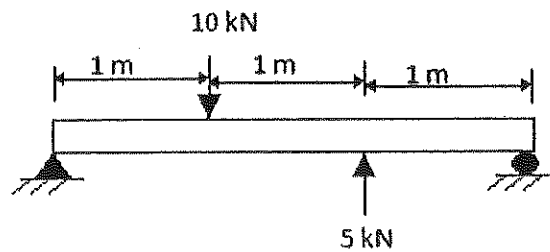
Now, putting $Q_1 = 0.04 \text{ m}^3/\text{s}$, $D_1 = d$, $D_2 = 2d$

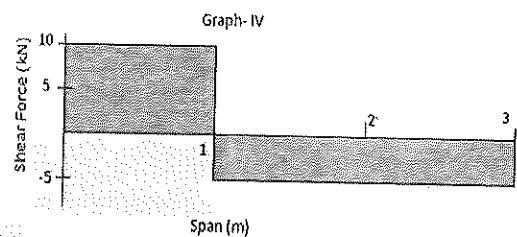
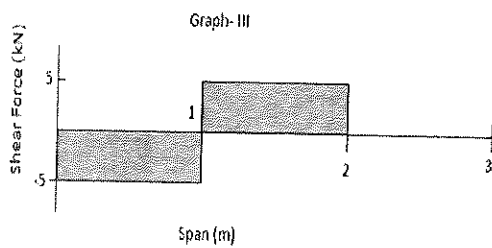
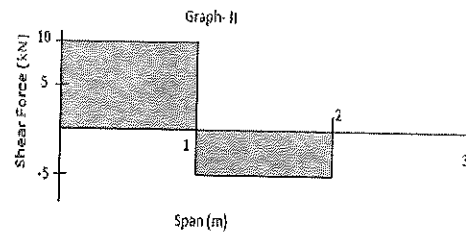
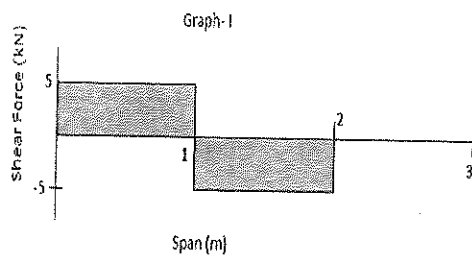
$$Q_2 = 0.04 \times \sqrt{\left(\frac{2d}{d}\right)^5}$$

$$Q_2 = 0.04 \times \sqrt{2^5}$$

$$Q_2 = 0.226 \text{ m}^3/\text{s}$$

Q 43. The shear force diagram for the shaft shown below resembles which one of the following graphs?





(A) Graph-I

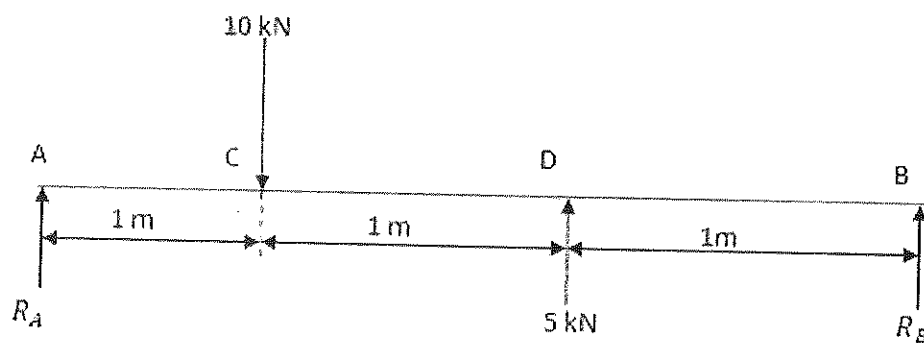
(B) Graph-II

(C) Graph-III

(D) Graph-IV

Answer (A)

Solution .



$$\text{Now, } \sum M_B = 0$$

$$R_A \times 3 - 10 \times 2 + 5 \times 1 = 0$$

$$\therefore R_A = \frac{20 - 5}{3} = 5 \text{ kN}$$

$$\text{Now, } \sum V = 0$$

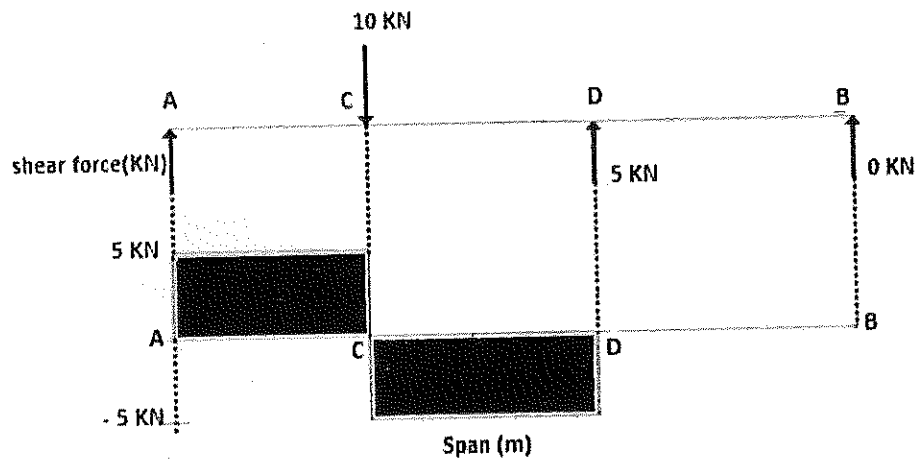
$$R_A + R_B - 10 + 5 = 0$$

$$R_B = 10 - 5 - 5 = 0$$

Now shear force calculation:

$$F_A = 5 \text{ KN} ; F_C = 5 - 10 = -5 \text{ KN}$$

$$F_D = 5 - 10 + 5 = 0, F_B = 0$$



Q 44. A 12 tonne diesel locomotive of 60 kW is employed in an underground haulage roadway. The coefficient of adhesion is 0.25 and the maximum gear efficiency is 80 %. The speed in m/s at which it will haul a train at its full power is

- (A) 2.548 (B) 2.448 (C) 2.038 (D) 1.630

Answer (D)

Solution :

We know,

$$\begin{aligned} \text{Tractive effort} &= \text{coefficient of adhesion} \times \text{mass} \\ &= 0.25 \times 12 \times 1000 \times 9.81 \\ &= 29430 \text{ N} \end{aligned}$$

Now,

$$\text{Power} = \frac{\text{tractive effort} \times \text{speed}}{\text{gear efficiency}}$$

$$\text{Or Speed} = \frac{\text{power} \times \text{gear efficiency}}{\text{tractive effort}}$$

$$\therefore \text{Speed} = \frac{60 \times 10^3 \times 0.80}{29430} \text{ m/s}$$

$$\therefore \text{Speed} = 1.630 \text{ m/s}$$

- Q 45. An air reservoir of volume 0.2 m^3 has an initial temperature of 27°C and pressure of 1800 kPa. After use, the air pressure falls to 1200 kPa at a temperature of 17°C . The volume of air consumed in m^3 corresponding to an air pressure of 101.3 kPa and temperature of 0°C is
- (A) 0.693 (B) 0.895 (C) 1.002 (D) 1.251

Answer (C)

Solution .

Let V is the volume of air used at 1800 kPa and 27°C

$\therefore (0.2 - V)$ volume of air remaining in the receiver.

$$\text{Here, } P_1 = 1800 \text{ kPa, } P_2 = 1200 \text{ kPa}$$

$$V_1 = (0.2 - V) \text{ m}^3, \quad V_2 = 0.2 \text{ m}^3$$

$$T_1 = 27 + 273 = 300 \text{ K}, \quad T_2 = 17 + 273 = 290 \text{ K}$$

We have,

$$\frac{P_1 V_1}{T_1} = \frac{P_2 V_2}{T_2}$$

$$\frac{1800 \times (0.2 - V)}{300} = \frac{1200 \times 0.2}{290}$$

$$\therefore V = 0.0620 \text{ m}^3$$

$$\text{Now, } \frac{P V}{T} = \frac{P_s V_s}{T_s}$$

$$\text{Here, } P = 1800 \text{ kPa; } P_s = 101.3 \text{ kPa}$$

$$T = 300 \text{ K; } T_s = 273 \text{ K}$$

$$V = 0.0620 \text{ m}^3; \quad V_s = ?$$

$$\therefore \frac{180 \times 0.0620}{300} = \frac{101.3 \times V_s}{273}$$

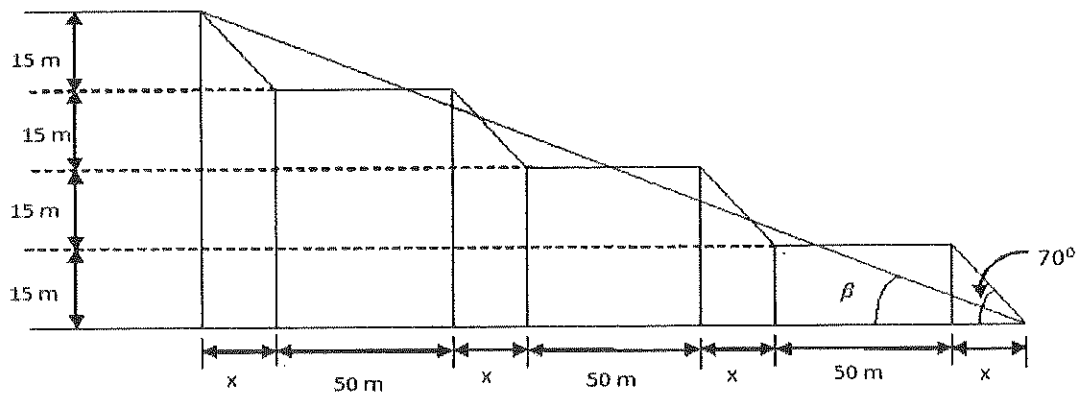
$$\therefore V_s = 1.002 \text{ m}^3$$

Q 46. Four benches are being worked by the opencast mining system. Height, width and face angle for each bench are 15 m, 50 m and 70° respectively. The overall slope angles of the benches in degrees is

- (A) 15.45 (B) 19.25 (C) 32.65 (D) 36.25

Answer (B)

Solution .



Where, β is overall slope angle

$$\text{Now, } \tan 70^\circ = \frac{H}{x}$$

$$\text{Or } x = \frac{15}{\tan 70^\circ} = 5.46 \text{ m}$$

$$\text{And } \tan \beta = \frac{15 \times 4}{5.46 \times 4 + 50 \times 3}$$

$$\therefore \beta = 19.25^\circ$$

Q 47. Match the following

Rock mass condition	Shaft sinking method	Limiting depth (m)
P. Water bearing strata of Loose sand or gravel	I. Freezing	1. 40
Q. Competent rock with Fissures and cracks filled With water	J. Depression of ground water level	2. 150
R. highly permeable coarse Solid or gravel with heavy water flow	K. Cement grouting	3. 1000
S. All types of water bearing rocks	L. Caisson	4. > 600
(A) P-L-4, Q-K-1, R-J-2, S-I-3	(B) P-L-1, Q-K-4, R-J-2, S-I-3	
(C) P-L-2, Q-K-4, R-J-3, S-I-1	(D) P-L-4, Q-K-3, R-J-2, S-I-1	

Answer (B)

Q 48. Match the following

System	device/Safety device
P. Drum winding	1. Taper guide
Q. Koepe winding	2. Detaching safety hook
R. Inclined Haulage	3. Rider
S. Winding in sinking shaft	4. Back catch
(A) P-1, Q-2, R-3, S-4	(B) P-4, Q-3, R-1, S-2

(C) P-2, Q-1, R-3, S-4

(D) P-2, Q-1, R-4, S-3

Answer (D)

Q 49. A closed container with 10 kg of air at ambient pressure and specific heat 1020 J/kg °C is cooled from 35°C. If the removal of 200 kJ of heat resulted in the saturation of air, the corresponding dew point temperature in °C is

(A) 33.0

(B) 27.3

(C) 15.4

(D) 12.9

Answer (C)

Solution .

We have,

$$H = Q.S.\Delta t$$

$$\text{And given that, } H = 200\text{kJ} = 200 \times 10^3 \text{ J}$$

$$Q = 10 \text{ kg}$$

$$S = 1024 \text{ J/kg}^\circ\text{C}$$

So,

$$200 \times 10^3 = 10 \times 1024 \times \Delta t$$

$$\therefore \Delta t = \frac{200 \times 10^3}{10 \times 1024}$$

$$\therefore \Delta t = 19.6^\circ\text{C}$$

$\therefore$ dew point temperature(T) is :

$$T = 35^\circ\text{C} - 19.6^\circ\text{C}$$

$$T = 15.4^\circ\text{C}$$

Q 50. Identify the INCORRECT statement

(A) Evasee is meant to minimise exit shock loss

(B) Evasee efficiency is primarily a function of divergence angle and area ratio

(C) Evasee produces an inevitable in friction losses

(D) Evasee installation leads to reduction in the fan total pressure

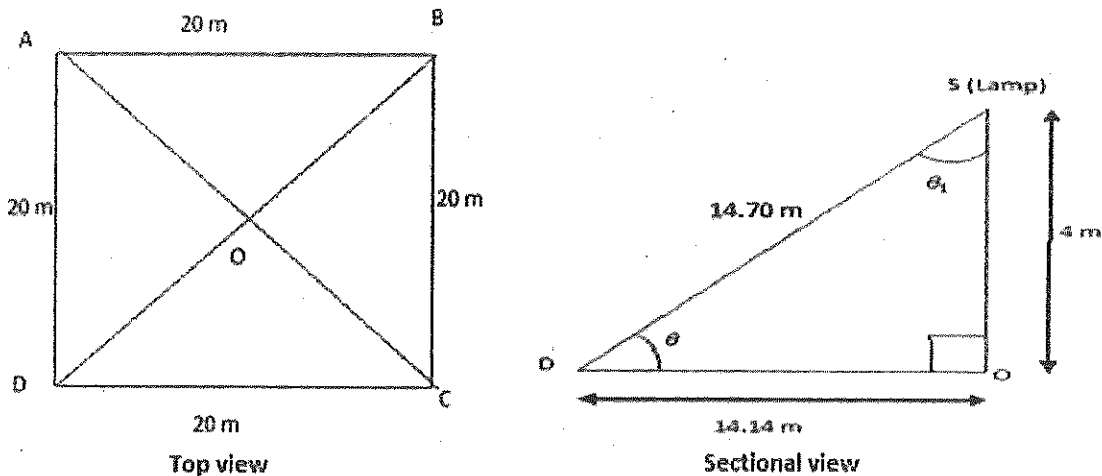
Answer (D)

Q 51. A single lamp placed centrally at the roof provides 40 lux illumination vertically below, at the floor of an underground workshop. The workshop is of dimensions $20\text{ m} \times 20\text{ m}$ with height 4 m . Assuming uniform spherical dispersion of luminous intensity, the floor level illumination in lux at any corner of the workshop is

(A) 23.2 (B) 10.9 (C) 3.0 (D) 0.8

Answer (D)

Solution .



By calculation, we find that,

$$BD = 28.29\text{ m}, DO = OB = 14.14\text{ m}$$

$$\text{And } \theta = \tan^{-1} \frac{4}{14.14} = 15.80^\circ$$

$$\text{Also, } \theta_1 = 180^\circ - 15.80^\circ - 90^\circ = 74.25^\circ$$

We know,

$$\text{Illumination at point} = (\text{candle of source}) \times \frac{\cos \theta_1}{(\text{distance})^2}$$

$$\text{Here, candle of source} = 40 \text{ lux} = 40 \times 4 \times 4 = 640 \text{ lumens}$$

$$\therefore \text{illumination at corner point D is} = \frac{640 \times \cos 74.25^\circ}{14.70^2} = 0.8 \text{ lux}$$

Q 52. An effluent sample is diluted with fresh water to make a solution of 300 ml. The DO of the solution initially is 8.0 mg/l and the value falls to 3.0 mg/l after 5 days. If the 5-day BOD of the original effluent is known to be 50 mg/l, the amount of fresh water added in ml to the solution is

- (A) 270 (B) 160 (C) 54 (D) 30

Answer (A)

Solution .

We know,

$$\text{Sample BOD in mg/L} = \frac{(\text{D.O. Depletion in mg/l}) \times (\text{Volume Diluted in ml})}{\text{Sample Volume in ml}}$$

$$\therefore \text{Sample Volume (V)} = \frac{(8 - 3) \times 300}{50}, \text{ml}$$

$$\therefore V = 30 \text{ ml}$$

Therefore ,

$$\text{Amount of fresh water (V')} \text{ added to the solution in ml} = (300 - 30) \text{ ml}$$

$$V' = 270 \text{ ml}$$

Q 53. With respect to stack emission the phenomenon of fumigation is noticed in case of

- (A) Atmospheric lapse rate being lower than the adiabatic lapse rate
- (B) Atmospheric lapse rate being higher than the adiabatic lapse rate
- (C) Temperature inversion in the atmosphere above the stack height
- (D) Temperature inversion in the atmosphere below the stack height

Answer (C)

Q 54. At a fan pressure of 450 Pa , 50 m³/s of air flows through a mine. When fan stops, 10 m³/s of air still flows in the same direction. The mine resistance in N s² m⁻⁸ is

- (A) 0.1731 (B) 0.1800 (C) 0.1875 (D) 0.2372

Answer (C)

Solution .

When fan running

$$(450 + N) = R \times 50^2 \quad [P = RQ^2]$$

$$\text{Or } R = \frac{450 + N}{50^2} \dots\dots\dots(1)$$

And when fan stopped

$$N = R_1 \times 10^2 \quad [N = N.V.P]$$

$$\text{Or } R_1 = \frac{N}{10^2} \dots\dots\dots(2)$$

Resistance of the mine is same whether fan is running or stopped

$$\therefore R = R_1$$

$$\frac{450 + N}{50^2} = \frac{N}{10^2}$$

$$\therefore N = 18.75 \text{ Pa}$$

From equation (1)

$$R = \frac{450 + 18.75}{2500}$$

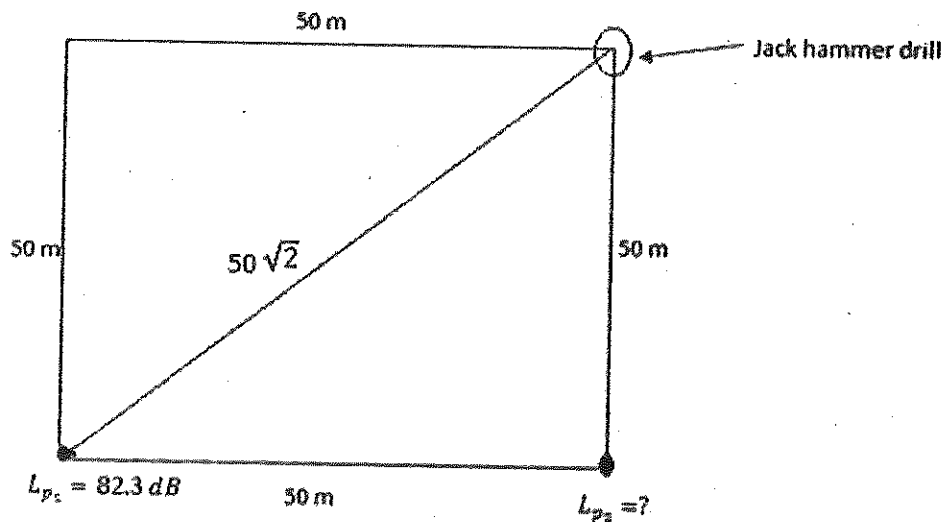
$$R = 0.1875 \text{ N s}^2 \text{ m}^{-8}$$

Q 55. A jackhammer operates at a corner of a square field of side 50 m. At the diagonally opposite corner, the SPL sensed is 82.3 dB. The SPL at any of the other two corners of the field in dB is

- (A) 86.3 (B) 85.3 (C) 83.6 (D) 81.2

Answer (B)

Solution .



If the sound pressure level, L_{p_1} , in dB is known at a distance r_1 (m) from a point source, the sound pressure level, L_{p_2} , in dB at a distance r_2 (m) from the same source can be calculated using the relationship :-

$$L_{p_2} = L_{p_1} - 20 \log \left(\frac{r_2}{r_1} \right)$$

$$\therefore L_{p_2} = 82.3 - 20 \log \left(\frac{50}{50\sqrt{2}} \right)$$

$$\therefore L_{p_2} = 85.3 \text{ dB}$$

Q 56. Consider the following data for the grade of iron ore from a working bench over past 5 weeks

Week	1	2	3	4	5
Grade (%)	62.1	61.0	60.5	62.5	62.0

The 3 - week moving average forecast for the grade , in % Fe , in the 6th week is

- (A) 61.66 (B) 61.90 (c) 62.20 (D) 62.50

Answer (A)

Solution .

The 3- week moving average forecast for the grade (G), in the 6th week is:

$$G = \frac{0.62 + 0.625 + 0.605}{3}$$

$$G = 0.6166$$

$$G = 61.66 \%$$

Q 57. The random variable X has the following probability mass fnction

$$P(4) = \frac{1}{4}, P(8) = \frac{1}{4}, P(12) = \frac{1}{4}, P(16) = \frac{1}{4}$$

The expected value of X

- (A) 1 (B) 3 (C) 10 (D) 12

Answer (C)

Solution .

Suppose random variable X can take value x_1 with probability P_1 , value x_2 with probability P_2 and so on, then expectation of random variable X is defined as –

$$E[X] = x_1 P_1 + x_2 P_2 + x_3 P_3 + \dots$$

$$\therefore E[X] = 4 \times \frac{1}{4} + 8 \times \frac{1}{4} + 12 \times \frac{1}{4} + 16 \times \frac{1}{4}$$

$$\therefore E[X] = 10$$

Q 58. The time between successive failure (in hours) of a side discharge loader operating in a mechanized underground coal mine are as follow

62 , 58 , 54 , 50 , 52 , 60 , 58 , 57 , 50 , 53

If the failure data follows an exponential distribution, then reliability of the equipment for a period of 50 hours s

- (A) 0.25 (B) 0.40 (C) 0.60 (D) 1.00

Answer (B)

Solution .

$$\text{Mean time between failure (m)} = \frac{62 + 58 + 54 + 50 + 52 + 60 + 58 + 57 + 50 + 53}{10}$$

$$m = 55.4 \text{ hours}$$

Reliability of the equipment run for a period of t hours is given by:

$$R(t) = e^{\frac{-t}{m}}$$

Here, t = 50 hours and m = 55.4 hours

$$\therefore R(t) = e^{\frac{-50}{55.4}}$$

$$\therefore R(t) = 0.40$$

Q 59. Three jobs A,B and C are to be assigned to three machines X, Y and Z. The processing costs are given below

		Machine		
		X	Y	Z
Job	A	19	28	31
	B	11	17	16
	C	12	15	13

The minimum cost of assigning the jobs to the machine is

- (A) 60 (B) 54 (C) 51 (D) 49

Answer (D)

Solution .

		Machine		
		X	Y	Z
Job	A	0	9	12
	B	0	6	5
	C	0	3	1

		Machine		
		X	Y	Z
Job	A	0	6	11
	B	0	3	4
	C	0	11	0

		Machine		
		X	Y	Z
Job	A	0	3	8
	B	0	0	1
	C	0	0	0

∴ Minimum total cost of assigning the job to the machine is : $19 + 17 + 13 = 49$

Q 60 An underground coal mine employing 1200 persons experienced 12 roof fall injuries during the year 2005. The roof fall injury rate per 1000 persons employed during the period 2005, as per the DGMS norms, is

- (A) 6 (B) 8 (C) 10 (D) 12

Answer (C)

Solution.

$$\begin{aligned} \text{Roof fall injury rate per 1000 persons employed} &= \frac{12}{1200} \times 1000 \\ &= 10 \end{aligned}$$

Q 61. Consider the following linear programming problem:

$$\text{Maximize } Z = 6X_1 + 4X_2$$

$$\text{Subject to } 2X_1 \leq 8$$

$$2X_2 \leq 12$$

$$3X_1 + 2X_2 \leq 18$$

$$X_1 \geq 0, X_2 \geq 0$$

The multiple optimal solutions lie on the line joining the corner points

(A) (0, 0), (0, 6) (B) (0, 6), (2, 6) (C) (2, 6), (4, 3) (D) (4, 3), (4, 0)

Answer (C)

Solution .

Multiple optimal solution – linear programming having more than one such point give maximum value among all remaining points.

Here, points (2, 6), (4, 3) give multiple optimal solution i.e.

$$\text{When (2,6), } Z_1 = 6 \times 2 + 4 \times 6 = 36$$

$$\text{And when (4,3), } Z_2 = 6 \times 4 + 4 \times 2 = 36$$

$$\therefore Z_1 = Z_2 = 36$$

Q 62. . Match the following

Problem	Technique
P. Queueing	1. Time series models
Q. Project scheduling and monitoring	2. Linear programming models
R. Transportation	3. Waiting line models
S. Forecasting of production	4. PERT and CPM
(A) P-3, Q-4, R-2, S-1	(B) P-2, Q-3, R-4, S-1

(C) P-3, Q-4, R-1, S-2

(D) P-2, Q-4, R-3, S-1

Answer (A)

Q 63. The net present value in Rs. of a 3 – year annuity of Rs. 10,000 discounted at 10% is

(A) 9,091

(B) 17,355

(C) 24,869

(D) 26,446

Answer (C)

Solution .

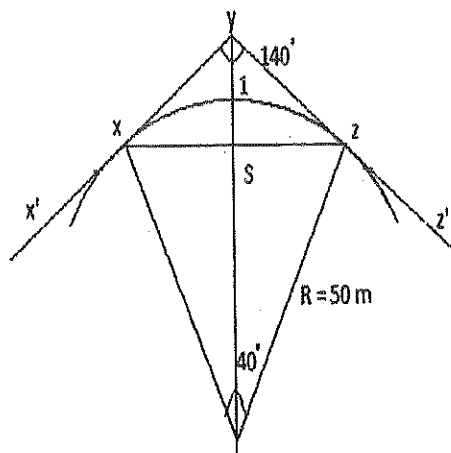
We have,

$$\text{Net present value} = \frac{R_1}{(1+i)} + \frac{R_2}{(1+i)^2} + \dots + \frac{R_n}{(1+i)^n}$$

$$\text{NPV} = \frac{10000}{(1+0.1)} + \frac{10000}{(1+0.1)^2} + \frac{10000}{(1+0.1)^3}$$

$$\text{NPV} = 24869 \text{ rupees}$$

Q 64. For a track gauge of 1.05 m and a speed of 10 km/hour, the super-elevation in cm from the following figure is



(A) 1.65

(B) 2.76

(C) 5.54

(D) 6.64

Answer (A)

Solution .

We have,

$$\text{Super elevation (E)} = \frac{a.v^2}{g.r} \text{ m}$$

Given that, $a = 1.05 \text{ m}$, $v = 10 \text{ km} = 2.8 \text{ m/s}$,

$$r = 50 \text{ m}, g = 9.81 \text{ m/s}^2$$

$$E = \frac{1.05 \times 2.8^2}{9.81 \times 50}, \text{ m}$$

$$E = 0.0167 \text{ m}$$

$$E = 1.67 \text{ cm}$$

- Q 65. In the bubble tube of a dumpy level, the bubble moves 5 mm for a change of inclination of $40''$. The sensitivity in mm and the radius of the bubble tube in m are (1 radian = $206265''$)
- (A) 0.125, 12.89 (B) 0.063, 26.78 (C) 0.125, 25.78 (D) 0.063, 12.89

Answer (C)

Solution .

$$\text{Since, } 40'' = 5 \text{ mm}$$

$$1'' = \frac{5}{40} \text{ mm}$$

$$1'' = 0.125 \text{ mm}$$

Sensitivity of the dumpy level = 0.125 mm

And we have

$$\alpha = \frac{n}{R}$$

$$\text{Or } R = \frac{n}{\alpha}$$

$$R = \frac{5 \times 206265}{40}, \text{ mm}$$

$$R = \frac{5 \times 206265}{40 \times 1000}, \text{ mm}$$

$$\therefore R = 25.78 \text{ m}$$

Q 66. The value of $A.B$, if $A + B = \begin{bmatrix} 1 & -1 \\ 3 & 0 \end{bmatrix}$ and $A - B = \begin{bmatrix} 3 & 1 \\ 1 & 4 \end{bmatrix}$, is

(A) $-4 \begin{bmatrix} 1 & 1 \\ 0 & 3 \end{bmatrix}$ (B) $-2 \begin{bmatrix} 1 & 1 \\ 0 & 3 \end{bmatrix}$

(C) $\begin{bmatrix} 1 & 1 \\ 0 & 3 \end{bmatrix}$ (D) $-\frac{1}{2} \begin{bmatrix} 1 & 1 \\ 0 & 3 \end{bmatrix}$

Answer (B)

Solution .

$$A + B = \begin{bmatrix} 1 & -1 \\ -3 & 0 \end{bmatrix} \dots\dots\dots(1)$$

$$A - B = \begin{bmatrix} 3 & 1 \\ 1 & 4 \end{bmatrix} \dots\dots\dots(2)$$

Solving (1) & (2), it is found that

$$A = \begin{bmatrix} 2 & 0 \\ 2 & 2 \end{bmatrix} \quad \& \quad B = \begin{bmatrix} -1 & -1 \\ 1 & -2 \end{bmatrix}$$

Now, $A.B = \begin{bmatrix} 2 & 0 \\ 2 & 2 \end{bmatrix} \cdot \begin{bmatrix} -1 & -1 \\ 1 & -2 \end{bmatrix}$

$$A.B = \begin{bmatrix} -2 + 0 \times 1 & -2 + (-2) \times 0 \\ -2 + 2 & -2 - 4 \end{bmatrix}$$

$$\therefore A.B = -2 \begin{bmatrix} 1 & 1 \\ 0 & 3 \end{bmatrix}$$

Q 67. The values of $f(x)$ at x_0, x_1 and x_2 are 9.0, 12.0 and 15.0 respectively. Using the Simpson's 1/3 rule, the value of $\int_{x_0}^{x_1} f(x)$, considering an interval of 0.1 is

(A) 1.2 (B) 2.4 (C) 1.6 (D) 1.8

Answer (B)

Solution .

Simpson's 1/3 rule,

$$\int_{x_0}^{x_0+nh} f(x) dx = \frac{h}{3} [(y_0 + y_n) + 4(y_1 + y_3 + \dots + y_{n-1}) + 2(y_2 + y_4 + \dots + y_{n-2})]$$

Given that, $h = 0.1, x_0 = 9, x_1 = 12, x_2 = 15;$

$$\int_{x_0}^{x_2} f(x) dx = \frac{0.1}{3} [9 + 15 + 4 \times 12]$$

$$\therefore \int_{x_0}^{x_2} f(x) dx = 2.4$$

Q 68. From the following page of the leveling field book, the missing values in F.S. and B.S. respectively

Station	B.S.	I.S.	F.S.	Rise	Fall	Remarks
1.	4.550					Starting point
2.	2.125		?		0.750	Change point
3.		2.225				
4.	?		1.975			Change point
5.		2.445		1.500		

(A) 3.804, 0.945

(B) 3.804, 3.945

(C) 5.300, 0.945

(D) 5.300, 3.945

Answer (D)

Solution .

$\therefore$ Fore Sight = back sight + fall

$$= 4.550 + 0.750$$

$$= 5.300$$

And Back Sight = intermediate reading + rise

$$= 2.445 + 1.500$$

$$= 3.945$$

Station	B.S	I.S	F.S	Rise	Fall	Remarks
1.	4.550					starting point
2.	2.125		5.300		0.750	change point
3.		2.225				
4.	3.945		1.975			change point
5.		2.445		1.500		

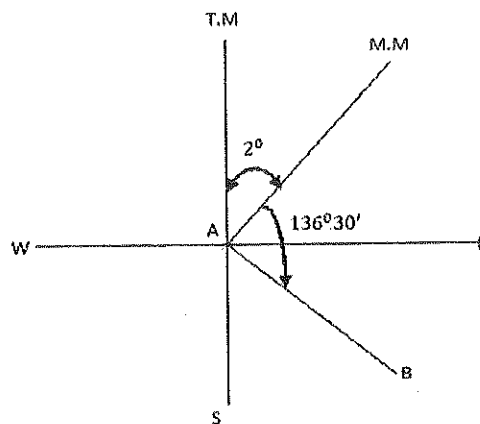
Q 69. The magnetic bearing and declination of a line were recorded in the year 1906 as $S48^{\circ}30'E$ and $2^{\circ}00'E$ respectively. If the declination in the year 2006 is $3^{\circ}00'W$, the magnetic bearing of the line is

- (A) $S48^{\circ}30'E$ (B) $S45^{\circ}30'E$ (C) $S41^{\circ}30'E$ (D) $S38^{\circ}30'E$

Answer (D)

Solution .

In the year 1906 -



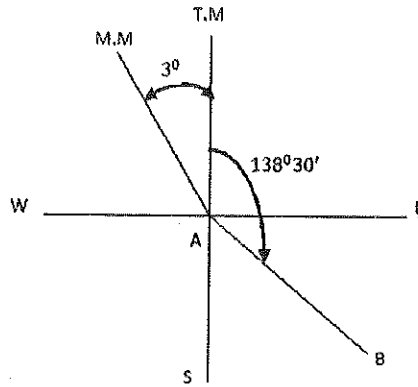
Magnetic bearing of line (AB) = $S 40^{\circ}30'E$

$$= 136^{\circ}30'0''$$

Declination = $2^{\circ}0'0''\text{E}$

True bearing of line (AB) = $136^{\circ}30'0'' + 2^{\circ}0'0'' = 138^{\circ}30'0''$

In the year 2006 -



Magnetic bearing of line (AB) = $138^{\circ}30'0'' + 3^{\circ} = 141^{\circ}30'0''$

= S $38^{\circ}30'$ E

Common Data for Questions 70, 71, 72 :-

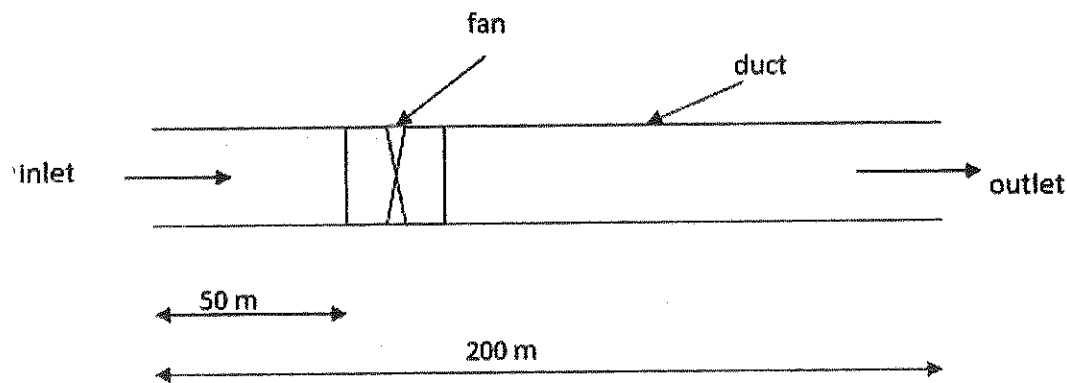
In a straight duct of length 200 m a fan operates 50 m away from the inlet such that the mean air velocity in the duct is 8 m/sec at a density of 1.1 kg/m^3 . The friction pressure loss per m length of the duct is 3.0 Pa and the entry shock factor is 1.2. Answer the following in terms of gauge pressure values in Pa

Q 70. The total pressure at the outlet of the duct is

- (A) -35.2 (B) 35.2 (C) 192.2 (D) 635.2

Answer (B)

Solution .



Total pressure at the out let of the duct = $P_T = P_v = \frac{1}{2} \rho v^2$

$$P_T = \frac{1}{2} \times 1.1 \times 8^2$$

$$\therefore P_T = 35.2 \text{ Pa}$$

Q 71. The total pressure at the inlet side of the fan is

- (A) -192.2 (B) -150.0 (C) 150.0 (D) 192.2

Answer (A)

Solution .

Total pressure(P') at inlet side of the fan = -[shock pressure + friction pressure]

$$P' = -[\text{shock factor} \times \text{velocity pressure} + \text{friction pressure}]$$

$$P' = -[1.2 \times 35.2 + 3 \times 50]$$

$$P' = -192.24 \text{ Pa}$$

Q 72. The total pressure generated by the fan is

- (A) 600 (B) 635.2 (C) 677.4 (D) 682.2

Answer (C)

Solution .

Total pressure (P'') generated by fan = velocity pressure + shock pressure + friction pressure

$$P'' = 35.2 + 1.2 \times 35.2 + 200 \times 3$$

$$P'' = 677.44 \text{ Pa}$$

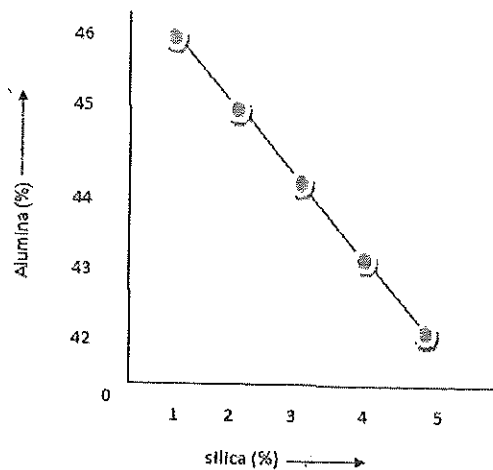
Common data for Questions 73,74 : A bauxite deposit has been intersected by 5 drill holes. The values of alumina (% by weight) and silica (% by weight) in these holes are as follows:

Drill hole number	Alumina (%)	Silica (%)
1	46	1
2	42	5
3	45	2
4	43	4
5	44	3

- Q 73. The relationship between alumina and silica is
- (A) positive linear (B) exponential
- (C) negative linear (D) random

Answer (C)

Solution.



∴ Relationship between alumina and silica is negative linear

Q 74. The unbiased estimate of variance of alumina and silica (%)² respectively are:

- (A) 2.5, 2.5 (B) 2.0, 2.5 (C) 2.5, 2.0 (D) 2.0, 2.0

Answer (D)

Solution .

We know,

Variance of random variable X is given by :

$$\text{Var } (X) = E[X^2] - (E[X])^2$$

For silica;

$$E[X^2] = \frac{1^2 + 5^2 + 2^2 + 4^2 + 3^2}{5} = 11$$

$$E[X] = \frac{1 + 5 + 2 + 4 + 3}{5} = 3$$

$$\therefore \text{Var } (X) = 11 - 3^2 = 2.0$$

For alumina;

$$E[X^2] = \frac{46^2 + 42^2 + 45^2 + 43^2 + 44^2}{5} = 1938$$

$$E[X] = \frac{46 + 42 + 45 + 43 + 44}{5} = 44$$

$$\therefore \text{Var } (X) = 1938 - 44^2 = 2.0$$

Statement for linked answer questions 75 & 76 : porosity of coarse grain sandstone

sample is 15 %. The specific gravity of sandstone is 2.8.

Q 75. What is the void ratio in the sandstone sample?

- (A) 0.150 (B) 0.176 (C) 0.850 (D) 1.176

Answer (B)

Solution 74.

We know,

$$\text{Porosity} = \frac{\text{void ratio}}{1 + \text{void ratio}}$$

$$0.15 = \frac{\text{void ratio}}{1 + \text{void ratio}}$$

$$\therefore \text{void ratio} = 0.176$$

Q 76. If the sandstone sample is fully saturated in water, the saturated density of sample in kg/m^3 is

(A) 1590

(B) 2234

(C) 2438

(D) 2531

Answer (D)

Solution .

We have ,

$$\text{Saturated density} = \frac{\text{mass of solid} + \text{mass of water}}{\text{volume of solid} + \text{volume of water}}$$

$$\text{Or Saturated density} = \frac{\text{volume of solid} \times \text{density of solid} + \text{density of water} \times \text{volume of water}}{\text{volume of solid} + \text{volume of water}}$$

Saturated density =

$$\frac{\frac{\text{volume of solid}}{\text{volume of solid}} \times \text{specific gravity of solid} \times \text{density of water} + \text{density of water} \times \frac{\text{volume of water}}{\text{volume of solid}}}{\frac{\text{volume of solid}}{\text{volume of solid}} + \frac{\text{volume of water}}{\text{volume of solid}}}$$

$$\therefore \text{Saturated density} = \frac{2.8 \times 1000 + 1000 \times \text{void ratio}}{1 + \text{void ratio}}$$

$$\text{Saturated density} = \frac{2800 + 1000 \times 0.176}{1 + 0.176}$$

$$\therefore \text{Saturated density} = 2531 \text{ kg/m}^3$$

Statement for linked answer questions 77 & 78 : A double outboard chain stranded conveyor is installed in an underground coal mine to transport coal. The mass of the chain and associated flight is 40 kg/m, the coefficient of kinematics friction are 0.33 between chain and the span and 0.5 between conveyed coal and the span. The motor efficiency is 80%. Coal is to be conveyed at the rate of 120 t/hour over a length of 120 m at a chain Speed of 0.9 m/s. The bulk density of coal is 900 kg/m³.

Q 77. The power requirement of the motor of the chain conveyor in kW is

- (A) 33.16 (B) 37.53 (C) 42.00 (D) 45.94

Answer (C)

Solution .

Power required to move empty belt :-

$$\therefore \text{Power (E)} = \text{force} \times \text{velocity}$$

$$\text{Or } E = \text{mass} \times \text{acceleration} \times \text{velocity}$$

$$\therefore E = 40 \times 120 \times 0.33 \times 9.81 \times 0.9$$

$$\therefore E = 13985 \text{ W} = 13.985 \text{ kW}$$

Power required to move load :-

$$H = \frac{120 \times 120 \times 1000 \times 9.81 \times 0.5}{60 \times 60}$$

$$\therefore H = 19620 \text{ W} = 19.620 \text{ kW}$$

$\therefore$ Power requirement (P) of motor of the chain conveyor :-

$$\therefore P = \frac{13.985 + 19.620}{0.80} = 42.00 \text{ kW}$$

Q 78 . The power requirement of the motor of the chain conveyor in kW , if it moves in the uphill direction at a gradient of 1 in 10, is

- (A) 46.91 (B) 42.00 (C) 38.53 (D) 30.16

Answer (A)

Solution .

Power required to move load vertically up :-

$$V = \frac{120 \times 1000 \times 9.81 \times 120 \times 1}{60 \times 60 \times 10}$$

$$V = 3924 \text{ W} = 3.924 \text{ kW}$$

∴ Power requirement of motor to convey in uphill:

$$P' = \frac{13.985 + 19.620 + 3.924}{0.80} = 46.91 \text{ kW}$$

Statement for linked answer questions 79 & 80 :-

The observed total time of drilling a face in an underground coal mine is 18 min. The rating of the drill screw performance, expressed in percentage is 90. Following allowances are recommended by the mine management

(i) Personal need allowance : 5% of the basic time

(ii) Fatigue allowance : 4% of the basic time

(iii) Contingency delay allowance : 1% of basic time

Q 79. The basic time required for drilling job by the crew in min is

- (A) 16.2 (B) 17.4 (C) 18.0 (D) 20.0

Answer (A)

Solution .

∴ Basic time (B.T) required for drilling Job = $18 \times 0.9 = 16.2$ minutes

Q 80. The standard time required for the same drilling job by the crew in min is

- (A) 15.50 (B) 17.01 (C) 17.82 (D) 18.90

Answer (C)

Solution.

Standard time (S.T) required for the same drilling Job :-

$$S.T = B.T + \frac{5}{100} \times (B.T) + \frac{4}{100} \times (B.T) + \frac{1}{100} \times (B.T)$$

$$S.T = 16.2 + \frac{5}{100} \times 16.2 + \frac{4}{100} \times 16.2 + \frac{1}{100} \times 16.2$$

$$\therefore S.T = 17.82 \text{ minutes}$$

Statement for linked answer Questions 81 & 82 : The result of a theodolite survey are

given below:

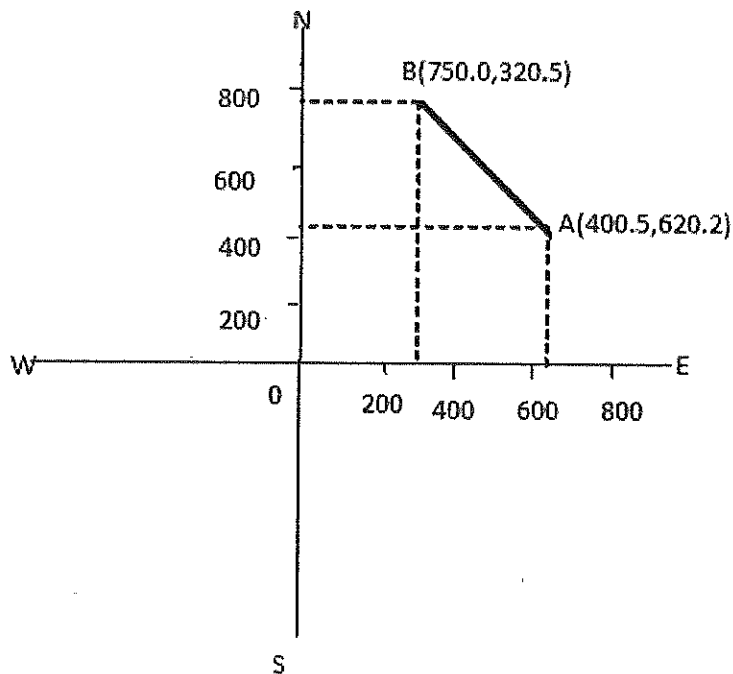
Points	North coordinate, in m	East coordinate, in m
A	400.5	620.2
B	750.5	320.5

Q 81. The length of line AB in m is

- (A) 460.78 (B) 349.70 (C) 106.60 (D) 50.30

Answer (A)

Solution .



$$\text{Latitude of line AB} = L_{AB} = 750.5 - 400.5 = 350 \text{ m}$$

$$\text{Departure of line AB} = D_{AB} = 620.2 - 320.5 = 299.7 \text{ m}$$

$$\begin{aligned} \therefore \text{Length of line AB} &= \sqrt{L_{AB}^2 + D_{AB}^2} \\ &= \sqrt{350^2 + 299.7^2} \\ &= 460.7 \text{ m} \end{aligned}$$

Q 82. The bearing of the line AB in degrees

- (A) 23.17 NE (B) 23.17NW (C) 40.57NW (D) 40.57NE

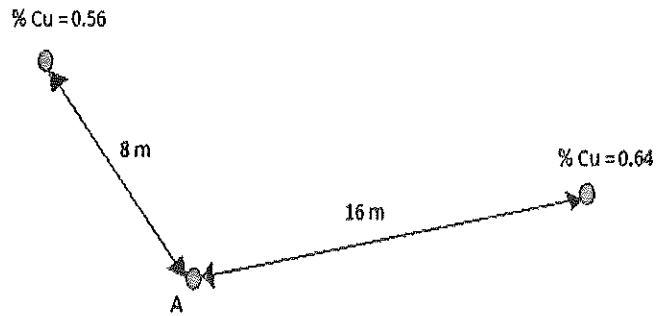
Answer (C)

Solution .

$$\begin{aligned} \text{Bearing of line AB} &= \tan^{-1} \left(\frac{\text{departure of AB}}{\text{latitude of AB}} \right) \\ &= \tan^{-1} \left(\frac{299.7}{350} \right) \\ &= 40.57^\circ \text{ NE} \end{aligned}$$

Statement for linked answer questions 83 & 84 :-

The following figure provides the grade information.



Q 83. The grade of copper (%) at a point A using inverse distance weighting method is

- (A) 0.47 (B) 0.58 (C) 0.61 (D) 1.20

Answer (B)

Solution .

Grade of copper at point A :-

$$g_A = \frac{\sum_{i=0}^n \frac{g_i}{d_i^2}}{\sum_{i=0}^n \frac{1}{d_i^2}}$$

$$g_A = \frac{\frac{0.56}{64} + \frac{0.64}{256}}{\frac{1}{64} + \frac{1}{256}}$$

$$g_A = 0.58 \%$$

Q 84. Assume the grade at A to be the average grade of the copper , mill recovery to be 85% and the smelting & refining losses to be 1.0 kg of copper per tonne of ore. The amount of saleable copper in kg/tonne of ore is

- (A) 2.93 (B) 3.93 (C) 4.93 (D) 5.93

Answer (B)

Solution .

$$\text{Copper content per tonne of ore} = \frac{0.58}{100} \times 1000 = 5.8 \text{ kg}$$

$$\text{Copper recovered by mill} = 5.8 \times \frac{85}{100} = 4.93 \text{ kg}$$

$$\therefore \text{Amount of saleable copper per tonne} = 5.8 - 1 = 4.8 \text{ kg}$$